

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

We now come to the promised cornerstone of this chapter. The basic problem is to determine conditions on an open set Ω that guarantee the existence of a conformal map $F : \Omega \rightarrow \mathbb{D}$.

A series of simple observations allow us to find necessary conditions on Ω . First, if $\Omega = \mathbb{C}$ there can be no conformal map $F : \Omega \rightarrow \mathbb{D}$, since by Liouville's theorem F would have to be a constant. Therefore, a necessary condition is to assume that $\Omega \neq \mathbb{C}$. Since $\mathbb{D}$ is connected, we must also impose the requirement that Ω be connected. There is still one more condition that is forced upon us: since $\mathbb{D}$ is simply connected, the same must be true of Ω (see Exercise 3). It is remarkable that these conditions on Ω are also sufficient to guarantee the existence of a biholomorphism from Ω to $\mathbb{D}$.

For brevity, we shall call a subset Ω of $\mathbb{C}$ **proper** if it is non-empty and not the whole of $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem) *Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \rightarrow \mathbb{D}$ such that*

$$F(z_0) = 0 \quad \text{and} \quad F'(z_0) > 0.$$

Corollary 3.2 *Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.*

Clearly, the corollary follows from the theorem, since we can use as an intermediate step the unit disc. Also, the uniqueness statement in the theorem is straightforward, since if F and G are conformal maps from Ω to $\mathbb{D}$ that satisfy these two conditions, then $H = F \circ G^{-1}$ is an automorphism of the disc that fixes the origin. Therefore $H(z) = e^{i\theta}z$, and since $H'(0) > 0$, we must have $e^{i\theta} = 1$, from which we conclude that $F = G$.

The rest of this section is devoted to the proof of the existence of the conformal map F . The idea of the proof is as follows. We consider all injective holomorphic maps $f : \Omega \rightarrow \mathbb{D}$ with $f(z_0) = 0$. From these we wish to choose an f so that its image fills out all of $\mathbb{D}$, and this can be achieved by making $f'(z_0)$ as large as possible. In doing this, we shall need to be able to extract f as a limit from a given sequence of functions. We turn to this point first.

3.2 Montel's theorem

Let Ω be an open subset of $\mathbb{C}$. A family $\mathcal{F}$ of holomorphic functions on Ω is said to be **normal** if every sequence in $\mathcal{F}$ has a subsequence that converges uniformly on every compact subset of Ω (the limit need not be in $\mathcal{F}$).

The proof that a family of functions is normal is, in practice, the consequence of two related properties, uniform boundedness and equicontinuity. These we shall now define.

The family $\mathcal{F}$ is said to be **uniformly bounded on compact subsets of Ω** if for each compact set $K \subset \Omega$ there exists $B > 0$, such that

$$|f(z)| \leq B \quad \text{for all } z \in K \text{ and } f \in \mathcal{F}.$$

Also, the family $\mathcal{F}$ is **equicontinuous** on a compact set K if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $z, w \in K$ and $|z - w| < \delta$, then

$$|f(z) - f(w)| < \epsilon \quad \text{for all } f \in \mathcal{F}.$$

Equicontinuity is a strong condition, which requires uniform continuity, uniformly in the family. For instance, any family of differentiable functions on $[0, 1]$ whose derivatives are uniformly bounded is equicontinuous. This follows directly from the mean value theorem. On the other hand, note that the family $\{f_n\}$ on $[0, 1]$ given by $f_n(x) = x^n$ is *not* equicontinuous since for any fixed $0 < x_0 < 1$ we have $|f_n(1) - f_n(x_0)| \rightarrow 1$ as n tends to infinity.

The theorem that follows puts together these new concepts and is an important ingredient in the proof of the Riemann mapping theorem.

Theorem 3.3 *Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω . Then:*

- (i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω .
- (ii) $\mathcal{F}$ is a normal family.

The theorem really consists of two separate parts. The first part says that $\mathcal{F}$ is equicontinuous under the assumption that $\mathcal{F}$ is a family of *holomorphic* functions that is uniformly bounded on compact subsets of Ω . The proof follows from an application of the Cauchy integral formula and hence relies on the fact that $\mathcal{F}$ consists of holomorphic functions. This conclusion is in sharp contrast with the real situation as illustrated by the family of functions given by $f_n(x) = \sin(nx)$ on $(0, 1)$, which is